GENERATION OF DIGITAL SIGNALS AND SPECTRAL ANALYSIS

PRE-LAB 1 · DIGITAL SIGNAL PROCESSING LABORATORY

OBJECTIVES

1. To generate common digital signals in MATLAB, namely Gaussian white noise and a sum of sinusoids, and to observe how each behaves in the frequency domain.
2. To estimate the bandwidth of these generated signals and of a recorded speech waveform using the discrete Fourier transform.

MATLAB PROGRAM

The whole exercise is carried out by a single script. Each part builds its signal, obtains the magnitude spectrum through the fast Fourier transform (*fft*), writes the figure to a PNG file, and prints the quantities asked for — frequency resolution, the verification of amplitude and phase, and the observed bandwidth — to the Command Window.

```
clc; clear; close all;

% ==========================================================================
% PRE-LAB 1 : Generation of Digital Signals & Signal Spectral Analysis
%
% Parts:
%   1. White (Gaussian) noise
%   2. Sum of three digital sinusoids
%   3. Multiplication of two digital sinusoids
%   4. Digital speech waveform (we.dat)
% Every figure is saved as a PNG (output_*.png) and every requested
% answer (frequency range, resolution, bandwidth, ...) is printed.
% ==========================================================================

matric = 'output';
fs = 8000;                  % common sampling rate (Hz)

%% =========== 1. WHITE NOISE WITH GAUSSIAN DISTRIBUTION ==================
x = 5*randn([1,4096]);        % 4096 Gaussian noise samples

Ak = abs(fft(x))/length(x);    % magnitude spectrum (DFT coefficients)
k  = 0:1:length(x)-1;          % frequency index
f  = k*fs/length(x);          % convert index to Hz

figure('Name','White Noise','Position',[100 100 700 500]);
subplot(2,1,1); stem(x(1:50));
title('White Gaussian Noise - first 50 samples');
xlabel('Sample index n'); ylabel('x(n)'); grid on;
subplot(2,1,2); plot(f,Ak);
title('Magnitude Spectrum of White Noise (f_s = 8000 Hz)');
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xlabel('Frequency (Hz)'); ylabel('|A_k|'); grid on;
print(gcf,[matric '_1_WhiteNoise'],'-dpng','-r200');

res_noise = fs/length(x);
fprintf('\n--- PART 1: White noise ---\n');
fprintf('Useful frequency range to read : 0 to fs/2 = %g Hz\n', fs/2);
fprintf('Frequency resolution          : fs/N = %.4f Hz\n', res_noise);
% sound(x/max(abs(x)),fs);     % (uncomment to listen)

%% ============ 2. SUM OF THREE DIGITAL SINUSOIDS ==========================
t = 0:(1/fs):0.5;             % time vector (step = 1/8000 s)
x1 = 5*cos(2*pi*500*t);
x2 = 5*cos(2*pi*1200*t + 0.25*pi);
x3 = 5*cos(2*pi*1800*t + 0.5*pi);

% (a) plot the three signals over ~0.01 s
figure('Name','Three Sinusoids','Position',[100 100 700 600]);
subplot(3,1,1); plot(t,x1); axis([0 0.01 -6 6]);
title('x_1 = 5cos(2\pi500t)'); xlabel('t (s)'); ylabel('x_1'); grid on;
subplot(3,1,2); plot(t,x2); axis([0 0.01 -6 6]);
title('x_2 = 5cos(2\pi1200t + 0.25\pi)'); xlabel('t (s)'); ylabel('x_2'); grid on;
subplot(3,1,3); plot(t,x3); axis([0 0.01 -6 6]);
title('x_3 = 5cos(2\pi1800t + 0.5\pi)'); xlabel('t (s)'); ylabel('x_3'); grid on;
print(gcf,[matric '_2a_ThreeSinusoids'],'-dpng','-r200');

fprintf('\n--- PART 2: Sum of sinusoids ---\n');
fprintf('Verify amplitudes / phases at t = 0:\n');
fprintf('  x1(0) measured = %.4f  (calc 5cos0        = %.4f)   x1max = %.4f\n', x1(1), 5*cos(0),       max(x1));
fprintf('  x2(0) measured = %.4f  (calc 5cos(0.25pi) = %.4f)   x2max = %.4f\n', x2(1), 5*cos(0.25*pi),  max(x2));
fprintf('  x3(0) measured = %.4f  (calc 5cos(0.5pi)  = %.4f)   x3max = %.4f\n', x3(1), 5*cos(0.5*pi),   max(x3));

% (b) sum of sinusoids
x = x1 + x2 + x3;
figure('Name','Sum of Sinusoids','Position',[100 100 700 500]);
subplot(2,1,1); plot(t,x); axis([0 0.01 -15 15]);
title('x(t) = x_1 + x_2 + x_3'); xlabel('t (s)'); ylabel('x'); grid on;

% (c) spectrum of the sum
Ak = abs(fft(x))/length(x);
k  = 0:1:length(x)-1;
f  = k*fs/length(x);
subplot(2,1,2); plot(f,Ak);
title('Spectrum of x(t) = x_1 + x_2 + x_3');
xlabel('Frequency (Hz)'); ylabel('|A_k|'); grid on;
print(gcf,[matric '_2bc_SumSpectrum'],'-dpng','-r200');

res_sum = fs/length(x);
fprintf('Sampling rate                : %g Hz\n', fs);
fprintf('Frequencies present in x(t)   : 500, 1200, 1800 Hz (+ mirror images at fs-f)\n');
fprintf('Maximum visible frequency     : fs/2 = %g Hz\n', fs/2);
fprintf('Frequency resolution          : fs/N = %.4f Hz\n', res_sum);

%% ============ 3. MULTIPLICATION OF TWO DIGITAL SINUSOIDS ================
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```
% x1 and x2 as defined above
fprintf('\n--- PART 3: Multiplication of sinusoids ---\n');
fprintf('  x1max = %.4f   x1rms = %.4f\n', max(x1), sqrt(mean(x1.*x1)));
fprintf('  x2max = %.4f   x2rms = %.4f\n', max(x2), sqrt(mean(x2.*x2)));

y = x1.*x2;                  % product signal

Ak = abs(fft(y))/length(y);
k  = 0:1:length(y)-1;
f  = k*fs/length(y);

figure('Name','Product of Sinusoids','Position',[100 100 700 700]);
subplot(4,1,1); plot(t,x1); axis([0 0.01 -6 6]);
title('x_1 = 5cos(2\pi500t)'); xlabel('t (s)'); ylabel('x_1'); grid on;
subplot(4,1,2); plot(t,x2); axis([0 0.01 -6 6]);
title('x_2 = 5cos(2\pi1200t + 0.25\pi)'); xlabel('t (s)'); ylabel('x_2'); grid on;
subplot(4,1,3); plot(t,y); axis([0 0.01 -15 15]);
title('y(t) = x_1(t) \times x_2(t)'); xlabel('t (s)'); ylabel('y'); grid on;
subplot(4,1,4); plot(f,Ak);
title('Spectrum of y(t)'); xlabel('Frequency (Hz)'); ylabel('|A_k|'); grid on;
print(gcf,[matric '_3_ProductSpectrum'],'-dpng','-r200');

res_prod = fs/length(y);
fprintf('Spectral peaks of y(t)        : 700 Hz and 1700 Hz (difference & sum freqs)\n');
fprintf('Amplitude of each peak        : (5*5)/2 = 12.5  -> ~%.3f in |A_k| (halved for one-sided)\n', 12.5/2);
fprintf('Frequency resolution          : fs/N = %.4f Hz\n', res_prod);

%% ============ 4. DIGITAL SPEECH WAVEFORM ================================
load we.dat;                  % loads variable "we"
x = we(:).';                  % ensure a row vector
n = 0:1:length(x)-1;
T = 1/fs;
tt = n*T;

Ak = abs(fft(x))/length(x);
k  = 0:1:length(x)-1;
f  = k*fs/length(Ak);

figure('Name','Speech Waveform','Position',[100 100 700 500]);
subplot(2,1,1); plot(tt,x);
title('Speech Waveform  we.dat  (f_s = 8000 Hz)');
xlabel('Time (s)'); ylabel('x(n)'); grid on;
subplot(2,1,2); plot(f,Ak);
title('Magnitude Spectrum of Speech Signal');
xlabel('Frequency (Hz)'); ylabel('|A_k|'); grid on;
print(gcf,[matric '_4_Speech'],'-dpng','-r200');

res_speech = fs/length(x);
fprintf('\n--- PART 4: Speech waveform ---\n');
fprintf('Number of speech samples      : %d\n', length(x));
fprintf('Approx. bandwidth (energy)    : most energy below ~1500-2000 Hz (read from plot)\n');
fprintf('Frequency resolution          : fs/N = %.4f Hz\n', res_speech);
% sound(x/max(abs(x)),fs);     % (uncomment to listen)
```

```
disp(' ');
disp('Done. All figures saved with the output prefix.');
```

PART 1: WHITE NOISE WITH A GAUSSIAN DISTRIBUTION

A block of 4096 normally distributed samples is created with *x = 5*randn([1,4096])*. The first fifty samples are drawn as a stem plot so the discrete nature of the sequence is visible, and the magnitude spectrum is then formed for a sampling frequency of f_s = 8000 Hz.

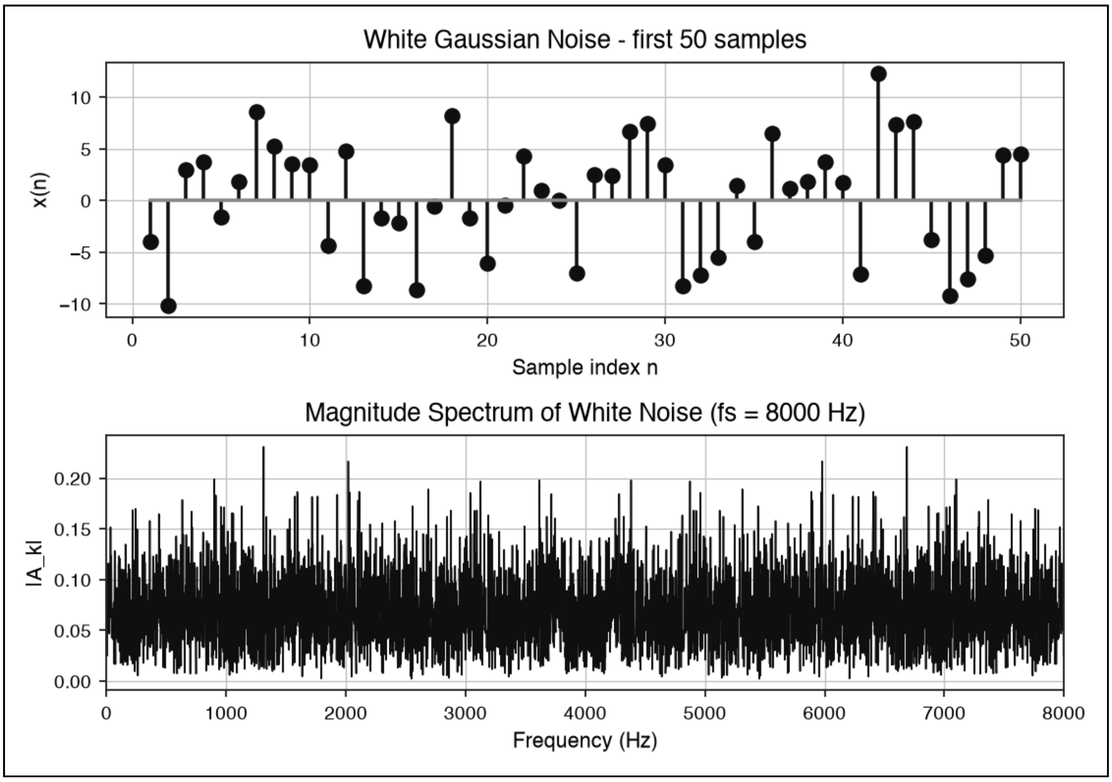

Figure 1: The first fifty noise samples (top) and the magnitude spectrum (bottom). The spectrum stays roughly level right across the band; this even spread of power over all frequencies is exactly what defines a white signal.

Quantity	Value
Useful frequency range to read	0 to $f_s/2$, that is 0 to 4000 Hz
Frequency resolution (8000 / 4096)	1.953 Hz

Table 1: Part 1 measurements.

Although the FFT is drawn over the full 0 to 8000 Hz axis, the signal is real, so the spectrum is mirror-symmetric about $f_s/2$. Only the lower half, 0 to 4000 Hz, holds independent information.

PART 2: SUM OF THREE DIGITAL SINUSOIDS

Three cosines are sampled with a step of 1/8000 s:

$$x_1(t) = 5\cos[2\pi(500)t]; \quad x_2(t) = 5\cos[2\pi(1200)t + 0.25\pi]; \quad x_3(t) = 5\cos[2\pi(1800)t + 0.5\pi]$$

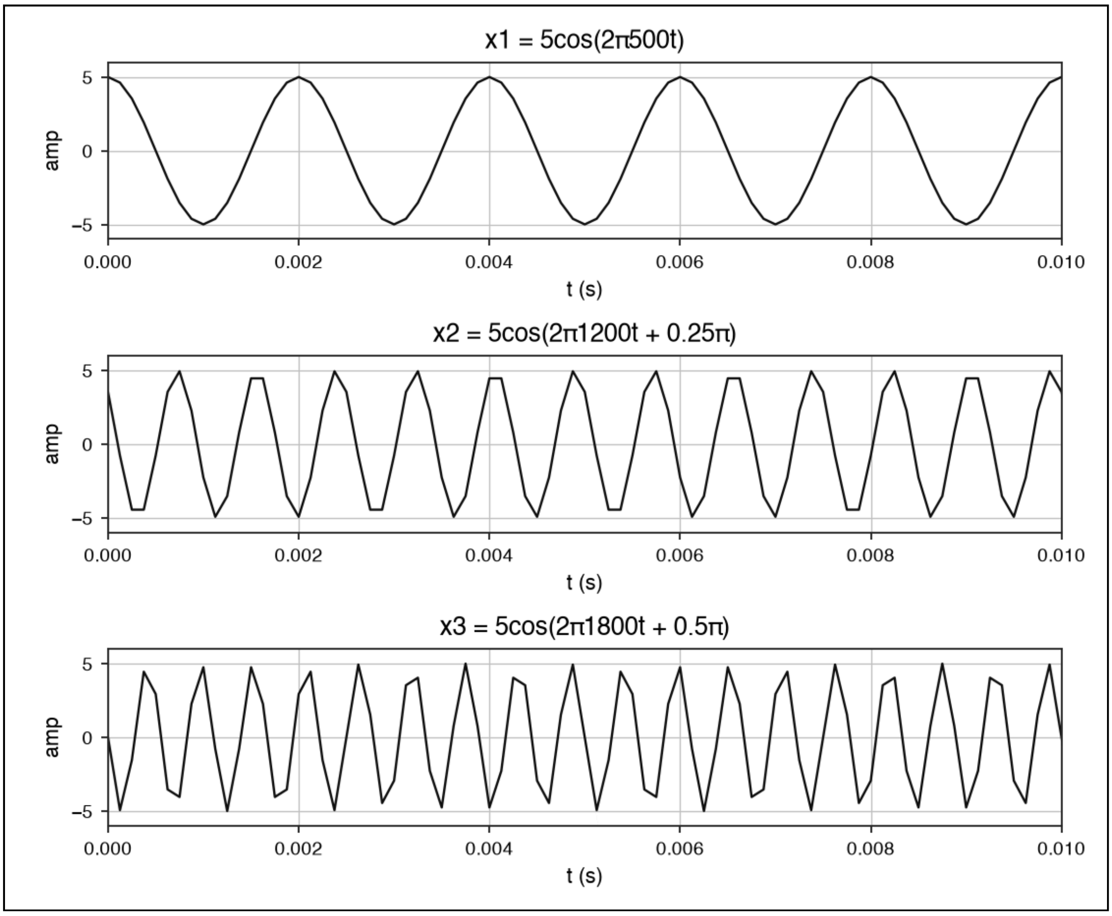

Figure 2: The three cosines over the first 0.01 s. Their starting values and peak heights agree with the amplitudes and phases that were set (see Table 2).

Signal	Calculated $x(0)$	Measured $x(0)$	Peak amplitude
x_1	$5\cos(0) = 5.000$	5.000	5.000
x_2	$5\cos(0.25\pi) = 3.536$	3.536	5.000
x_3	$5\cos(0.5\pi) \approx 0.000$	≈ 0.000	5.000

Table 2: Amplitude and phase check at t = 0.

Adding the three cosines gives $x(t) = x_1 + x_2 + x_3$, whose spectrum is then computed.

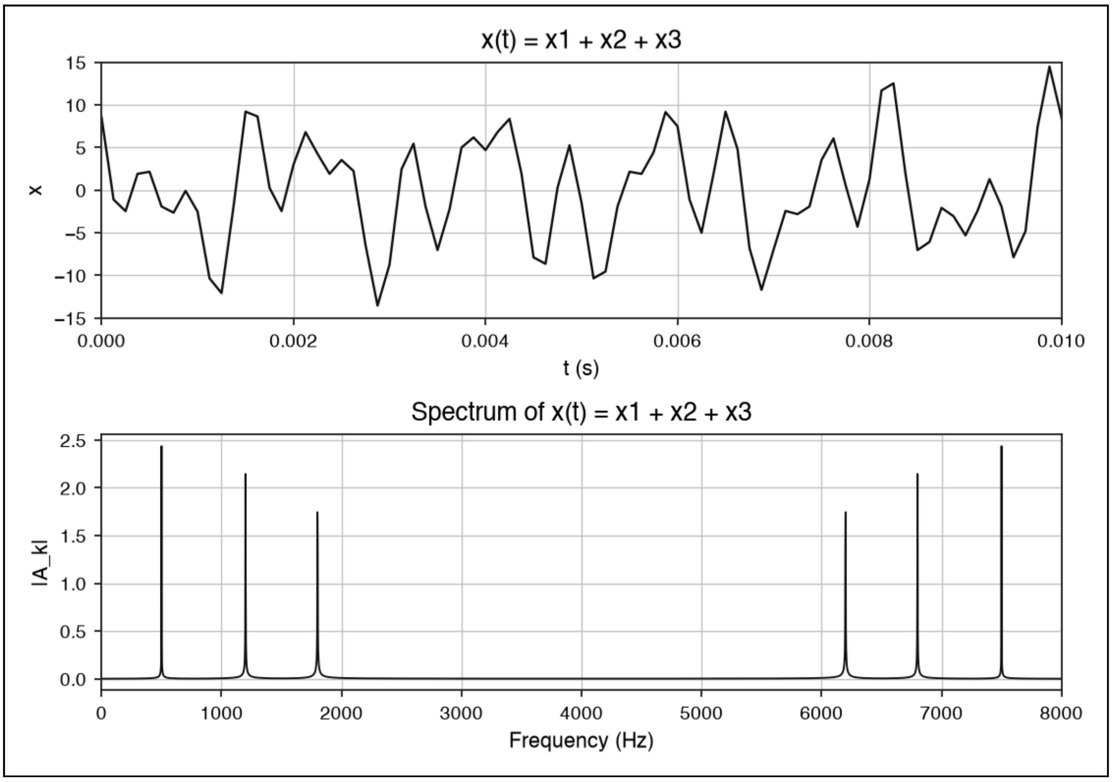

Figure 3: The combined signal (top) and its spectrum (bottom). Exactly three lines rise at 500, 1200 and 1800 Hz — the three component tones — and their mirror copies sit at 6200, 6800 and 7500 Hz.

Quantity	Value
Sampling rate	8000 Hz
Frequencies present in the spectrum	500, 1200, 1800 Hz (plus mirror images)
Highest visible frequency	$f_s/2$ = 4000 Hz
Frequency resolution	about 2.0 Hz

Table 3: Part 2 measurements.

PART 3: MULTIPLICATION OF TWO DIGITAL SINUSOIDS

Taking x_1 and x_2 from Part 2, their product $y(t) = x_1(t) \times x_2(t)$ is formed. The product-to-sum identity shows that multiplying two cosines produces one tone at the difference frequency and one at the sum frequency:

$$5\cos(2\pi 500t)\cdot 5\cos(2\pi 1200t+\varphi) = 12.5\cos(2\pi 700t-\varphi) + 12.5\cos(2\pi 1700t+\varphi)$$

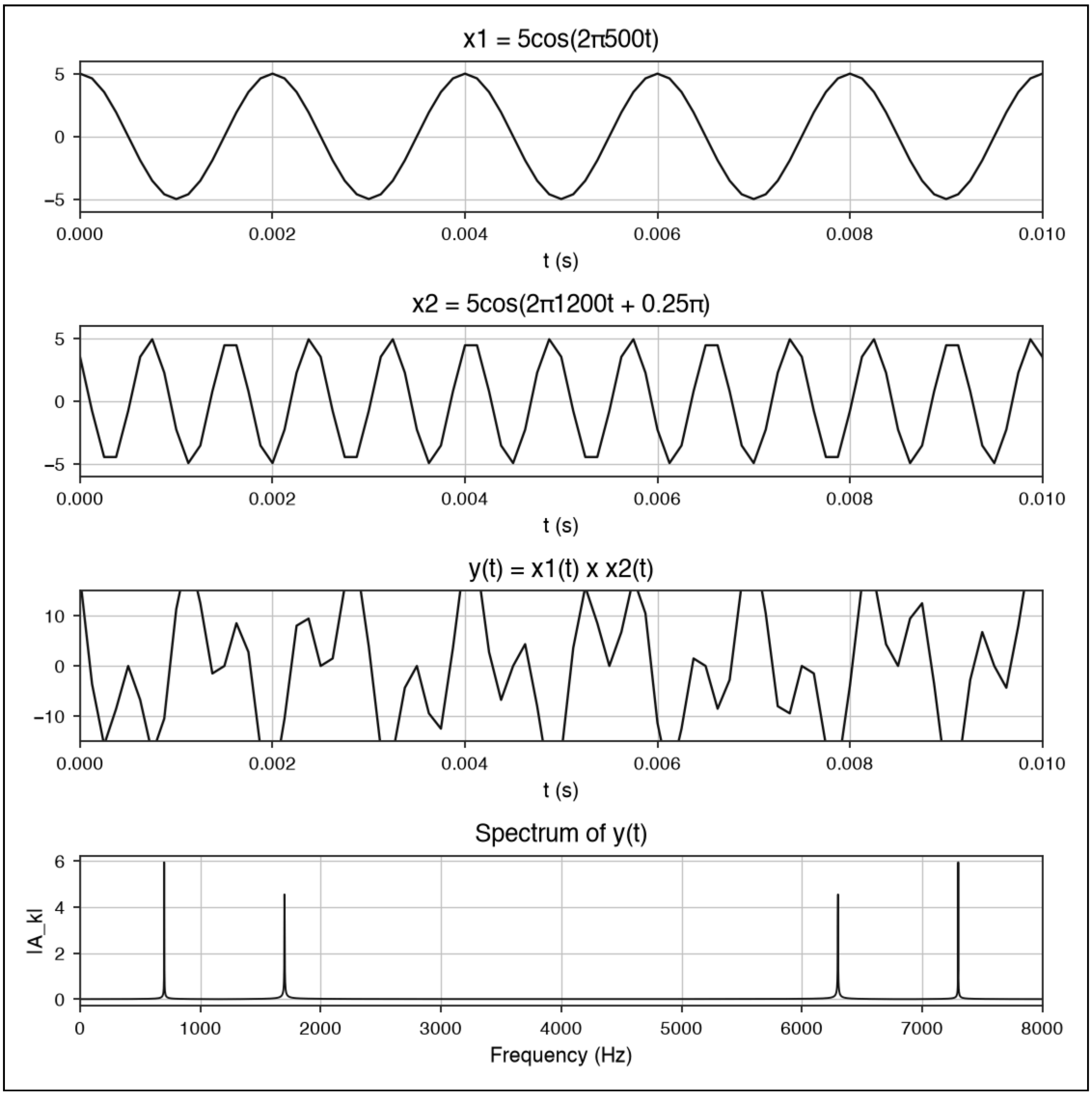

Figure 4: The two input tones, their product y(t), and the spectrum of y(t). Peaks appear only at the difference frequency (700 Hz) and the sum frequency (1700 Hz), with mirror copies at 6300 and 7300 Hz; the original 500 and 1200 Hz lines have disappeared.

Quantity	Value
x_1 peak and rms	5.000 and 3.536
x_2 peak and rms	5.000 and 3.536
Spectral peaks of y(t)	700 Hz and 1700 Hz
Peak amplitude (theory, $A_1A_2/2$)	12.5 (about 6.25 one-sided)
Frequency resolution	about 2.0 Hz

Table 4: Part 3 measurements.

PART 4: DIGITAL SPEECH WAVEFORM

The supplied speech record (*we.dat*, sampled at 8000 Hz) is read in, plotted against time and transformed to reveal where its energy lies.

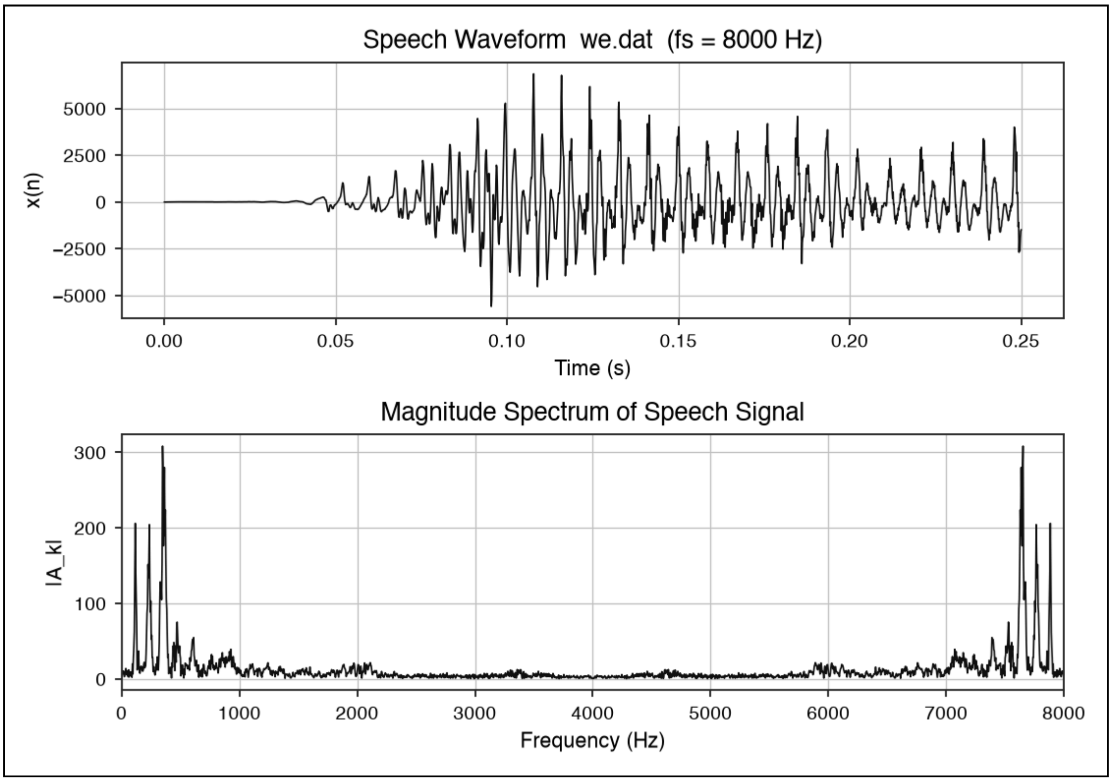

Figure 5: The speech waveform (top) and its magnitude spectrum (bottom). The energy is concentrated at low frequencies: the strongest content sits below roughly 600 Hz and has largely faded by about 1 kHz. The activity above 7 kHz is only the symmetric image of that low-frequency energy.

Quantity	Value
Number of samples (N)	2001
Approximate bandwidth	about 1 kHz (most energy below ~600 Hz)
Frequency resolution (8000 / 2001)	about 4.0 Hz

Table 5: Part 4 measurements.

CONCLUSION

Spectral analysis with the FFT brought out the frequency make-up of every test signal. The white noise gave a flat spectrum, the sum of sinusoids returned its three component tones, the product of two sinusoids left only the sum and difference frequencies, and the speech signal proved to be band-limited with most of its energy under about 1 kHz. Throughout, the resolution equals f_s/N, so lengthening the record (a larger N) sharpens the frequency detail.